ASYMPTOTIC SUCCESS UPPER BOUNDS FOR RESTRICTED CRT-LOCAL FACTORING CONTROLLERS AND ECM DUAL-CHAIN COUNTERFAMILIES

ARGUS OPEN RESEARCH PROJECT

Abstract. We prove two unconditional negative theorems against restricted deterministic integer-factoring controller models: (1) for one-gcd CRT-local controllers equipped with equality and Jacobi-symbol feedback, an infinite balanced-semiprime family limits asymptotic success to at most $5/8 + o(1)$; and (2) for the elliptic-curve dual-chain exponent model with resource parameters (c, e), a shell-uniform Rankin and prime-selection construction produces an infinite balanced-semiprime counterfamily with exponent parameter $\eta = 1/(256\max(3c + 2, e + 1))$.

1. Main Theorems

Theorem 1.1. *Let $N = pq$ range over balanced semiprimes.*

(1) *Every deterministic controller in the frozen one-gcd CRT-local grammar with equality and Jacobi-symbol branching succeeds on a fraction at most $5/8 + o(1)$ of balanced semiprimes as $N \to \infty$.*
(2) *For every fixed positive rational resource tuple (c, e) in the input-specific ECM dual-chain model, there exists an infinite family of balanced semiprimes $N = pq$ on which the dual-chain point multiples return to the identity O simultaneously across both prime factors without separating p and q.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1285 in the domain of computational number theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped lower bound theorem* deciding the scope: 5/8+o(1) success bound for one-gcd CRT-local controllers and balanced-semiprime counterfamily for frozen ECM dual-chain routes.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted classical polynomial-time integer factoring.

References

[1] Primary problem list and source literature for Result 1285 (computational number theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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